02_CRD_Analysis_non-normal_data

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CRD: Non-normal Data

1. Load Data

```
datacrd_non <- read.csv("crd_non_normal_data.csv")

head(datacrd_non)
```

```
  treatment yield_t_ha
1   Control       2.71
2   Control       2.85
3   Control       2.93
4   Control       3.02
5   Control       3.08
6   Control       3.15
```

---

2. Model Building

Initially, fit the same one-way ANOVA model used for the CRD.

```
modelcrd_non <- aov(
  yield_t_ha ~ treatment,
  data = datacrd_non
)

summary(modelcrd_non)
```

```
            Df Sum Sq Mean Sq F value   Pr(>F)
treatment    5   9.45  1.8899   8.494 5.54e-06 ***
Residuals   54  12.02  0.2225
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

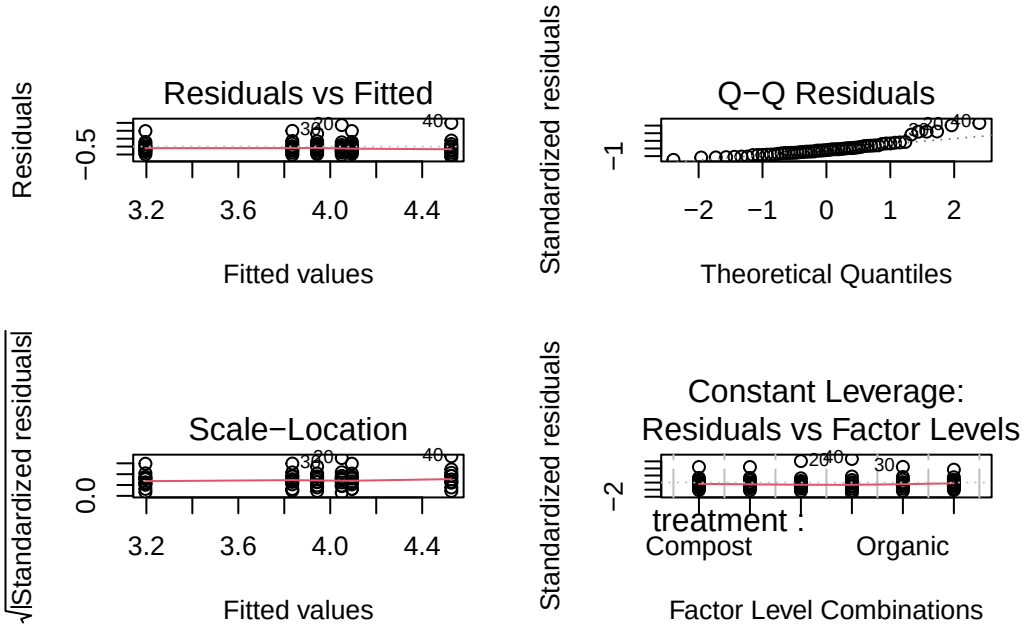
The ANOVA model is fitted initially so that its residuals can be diagnosed.

---

3. Model Diagnostics

Check the residuals to determine whether the ANOVA assumptions are reasonably satisfied.

```
par(mfrow = c(2, 2))
plot(modelcrd_non)
```




```
par(mfrow = c(1, 1))
```

Check residual normality:

```
shapiro.test(residuals(modelcrd_non))
```

```
    Shapiro-Wilk normality test

data:  residuals(modelcrd_non)
W = 0.86415, p-value = 8.179e-06
```

The diagnostic results indicate that the residuals do not reasonably satisfy the normality assumption.

Therefore, the ANOVA model is not appropriate for the primary analysis.

> **Decision:** Use the Kruskal-Wallis test as the non-parametric alternative for this CRD.

---

4. Kruskal-Wallis Test

```
kruskal.test(
  yield_t_ha ~ treatment,
  data = datacrd_non
)
```

```
    Kruskal-Wallis rank sum test

data:  yield_t_ha by treatment
Kruskal-Wallis chi-squared = 26.866, df = 5, p-value = 6.056e-05
```

Interpretation

- **H :** The treatment groups have the same distribution.
- **H :** At least one treatment group differs.

If $\mathbf{p < 0.05}$, reject H and conclude that there is evidence of a difference among the treatment groups.

If the test is significant, a post-hoc test is required to determine which groups differ.

---

5. Dunn's Post-hoc Test

```
library(FSA)
```

```
## FSA v0.10.0. See citation('FSA') if used in publication.
## Run fishR() for related website and fishR('IFAR') for related book.
```

```
dunn_result = dunnTest(
  yield_t_ha ~ treatment,
  data = datacrd_non,
  method = "bonferroni"
)
```

```
Warning: treatment was coerced to a factor.
```

```
dunn_result
```

```
Dunn (1964) Kruskal-Wallis multiple comparison

  p-values adjusted with the Bonferroni method.

              Comparison          Z      P.unadj        P.adj
1      Compost - Control  3.4636863 5.328274e-04 7.992411e-03
2          Compost - NPK  0.4353617 6.632999e-01 1.000000e+00
3          Control - NPK -3.0283247 2.459137e-03 3.688706e-02
4  Compost - NPK_Organic -1.4981564 1.340926e-01 1.000000e+00
5  Control - NPK_Organic -4.9618427 6.982753e-07 1.047413e-05
6      NPK - NPK_Organic -1.9335181 5.317241e-02 7.975861e-01
7      Compost - Organic  1.3444993 1.787870e-01 1.000000e+00
8      Control - Organic -2.1191870 3.407466e-02 5.111200e-01
9          NPK - Organic  0.9091376 3.632775e-01 1.000000e+00
10 NPK_Organic - Organic  2.8426557 4.473937e-03 6.710906e-02
11        Compost - Urea  0.7490782 4.538101e-01 1.000000e+00
12        Control - Urea -2.7146081 6.635423e-03 9.953135e-02
13            NPK - Urea  0.3137165 7.537364e-01 1.000000e+00
14    NPK_Organic - Urea  2.2472346 2.462504e-02 3.693756e-01
15        Organic - Urea -0.5954211 5.515620e-01 1.000000e+00
```

The Bonferroni-adjusted p-values are used to determine which treatment pairs differ significantly.

- **Adjusted p $<$ 0.05:** significant difference
- **Adjusted p 0.05:** no statistically significant difference detected

The final interpretation should identify **which treatments differ**, rather than simply reporting that the Kruskal-Wallis test was significant.

---

6. Treatment Grouping

Use the adjusted p-values to create letter groups:

```r
library(multcompView)
```

```
Warning: package 'multcompView' was built under R version 4.4.3
```

```r
p_values <- dunn_result$res$P.adj

names(p_values) <- gsub(
  " - ", "-",
  dunn_result$res$Comparison
)

letters <- multcompLetters(p_values)

letters$Letters
```

```
    Compost     Control         NPK NPK_Organic     Organic        Urea
        "a"         "b"         "a"         "a"        "ab"        "ab"
```

Treatments sharing the same letter are not significantly different; treatments with different letters are significantly different at the chosen significance level.